

CardioTrack

Sprint Goal : Extend CardioTrack's data model with an audit logging capability and a reference data table , applying full normalization , to move the schema clear to a production ready professional baseline.

Blocking board columns

Core already implemented :

- ☒ User
- ☒ Patient
- ☒ VitalSing
- ☒ VitalSignAlert
- ☒ Medication
- ☒ Appointment
- ☒ MedicalHistory
- ☒ RefreshToken
- ☒ EmailVerificationCode

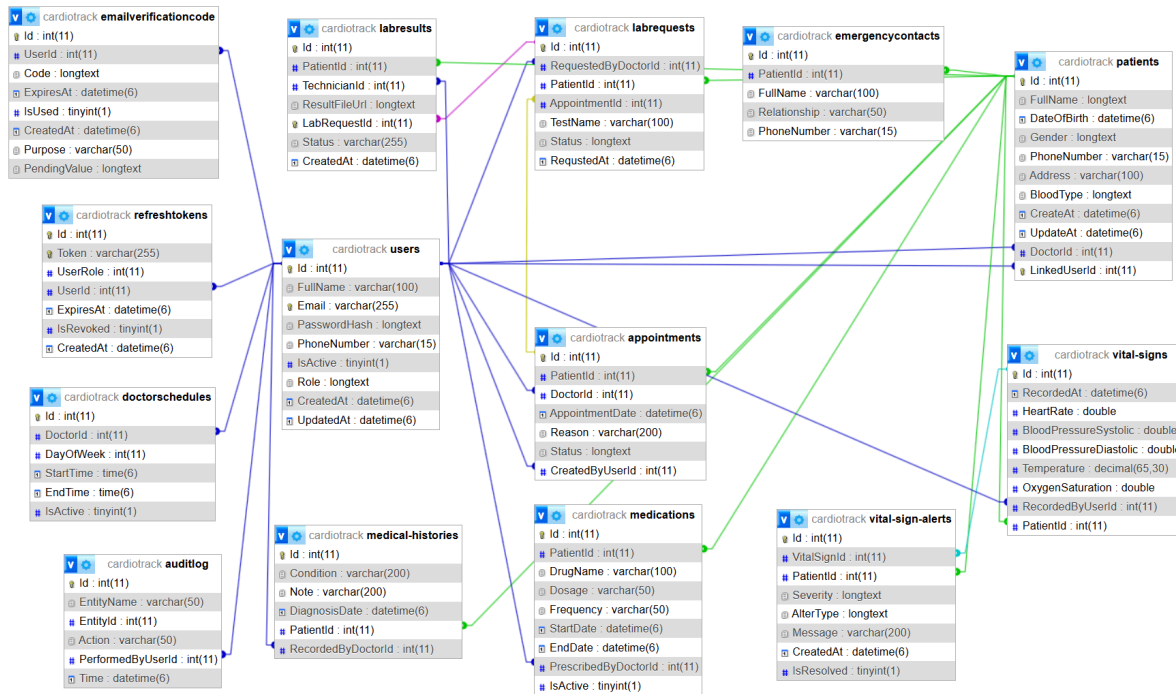
This Sprint :

- ☒ AuditLog

Future professional baseline :

- ☒ LapReques
- ☒ LapResult
- ☒ EmergencyContact

Database with all new entities



Sprint 1 Retrospective — CardioTrack

What went well

- Successfully extended the existing exception-handling and testing patterns from previous sprints to new features (AuditLog, Appointment transaction), keeping the codebase consistent instead of introducing new patterns.
- Reused existing domain concepts (VitalSignAlert, Appointment) instead of over-expanding the schema — avoided unnecessary complexity growth.
- Applied database transactions correctly to guarantee atomicity between related operations (appointment creation + alert resolution).

What to improve

- Some new entities (DoctorSchedule, LabRequest, LabResult, EmergencyContact) were scaffolded based on planning discussions before their actual model definitions were finalized and verified against the running codebase — led to assumptions that need re-verification.
- AuditLog logging was wired into only one service; broader adoption across sensitive operations (Medication, MedicalHistory) is still pending.
- New endpoints (Pagination/Filtering/Sorting, Prescription creation) lack dedicated unit or integration test coverage so far.

One concrete action for Sprint 2

Before adding any further new tables, finish and test-cover what was started this sprint: verify all new entity definitions against the actual codebase, extend AuditLog to at least two more services, and add test coverage for the Appointment transaction and Prescription stock-check logic.

Sprint 1 Retrospective — CardioTrack

What went well

- DoctorSchedule conflict detection logic correctly identifies overlapping time ranges using interval-overlap comparison logic.
- Schedule updates correctly cascade to affected appointments (auto-postpone), showing awareness of downstream effects when a doctor's availability changes.

What to improve

- The AuditLog integration was wired into DoctorScheduleService before the LogAsync call signature was double-checked against its interface — resulted in a type mismatch that needs fixing before this compiles cleanly.
- A reference-copy bug was introduced when trying to log "before" values in an update operation — a reminder that C# objects are reference types, and

snapshotting state requires an explicit copy, not just a new variable name.

- Placeholder-style exception messages (e.g. "Doctor.NotFound") were reintroduced despite an earlier decision to use clear plain-text messages.

One concrete action for Sprint 2

Before adding further features, do a full pass fixing the three issues above in DoctorScheduleService, then apply the same LogAsync signature check to every other service using AuditLog, to avoid the same mistake being repeated elsewhere in the codebase.

Sprint 1 Summary — CardioTrack

Sprint Goal

Extend CardioTrack's data model with an audit-logging capability and transactional business logic, while keeping the existing core workflow fully functional and consistent with established patterns.

What Was Delivered

- AuditLog entity + service, wired into account activation/deactivation
- Draft schema for DoctorSchedule, LabRequest, LabResult, EmergencyContact
- Paginated, filterable, sortable GET /Doctor/get-patients endpoint
- Transactional Appointment creation with availability checking, fee calculation, and optional linked-alert resolution

[Sprint 2] Fix AuditLog.LogAsync call signature in DoctorScheduleService (parameters currently mismatched — string passed where int/action expected)

[Sprint 2] Fix oldValue reference-copy bug in UpdateDoctorScheduleAsync — snapshot values before mutation, not after

[Sprint 2] Replace "Doctor.NotFound"/"Schedule.Conflict" placeholder-style messages with clear user-facing text

[Sprint 2] Reconsider postponing ALL same-day appointments on schedule

update

[Sprint 2] Extend AuditLog logging beyond DoctorSchedule to Medication, MedicalHistory, EmergencyContact, LabRequest, and LabResult services

[Sprint 2] Add unit/integration tests for new CRUD endpoints

(EmergencyContact, LabRequest, LabResult) and DoctorSchedule

[Sprint 2] Add Validator classes for the new request DTOs

(EmergencyContact, LabRequest, LabResult, DoctorSchedule)

Migration

id	name	safeName	applied
20260811010216_InitialCreate		InitialCreate	InitialCreate
True			
20260811202234_EditToken		EditToken	EditToken
True			
20260823081744_AddAuditLogTable		AddAuditLogTable	
AddAuditLogTable	True		
20260823082801_UpdateAnotation		UpdateAnotation	UpdateAnotation
True			
20260824171420_AddNewEntities		AddNewEntities	AddNewEntities
True			
20260824180652_SeedMedicationData		SeedMedicationData	
SeedMedicationData	True		
20260825083014_EditAuditLogEntity		EditAuditLogEntity	
EditAuditLogEntity	True		
20260826163843_AddMedicalOrder		AddMedicalOrder	
AddMedicalOrder	True		
20260826171250_DeleteAddMedicationEntity		DeleteAddMedicationEntity	
DeleteAddMedicationEntity	True		
20260826172037_AddFeeToAppointment		AddFeeToAppointment	
AddFeeToAppointment	True		